One-carbon metabolism and prostate cancer risk: prospective investigation of seven circulating B vitamins and metabolites.

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PURPOSE: Components of one-carbon metabolism are believed to influence cancer development with suggested mechanisms, including DNA methylation and DNA repair mechanisms. However, few prospective studies have investigated one-carbon metabolism in relation to prostate cancer risk, and the results have been conflicting. The aim of this study was to do a comprehensive investigation of the components of one-carbon metabolism in relation to prostate cancer risk. A panel of seven circulating B vitamins and related metabolites was selected, most of which have not been studied before. MATERIALS AND METHODS: We analyzed plasma concentrations of betaine, choline, cysteine, methionine, methylmalonic acid (MMA), vitamin B2, and vitamin B6 in 561 cases and 1,034 controls matched for age and recruitment date, nested within the population-based Northern Sweden Health and Disease Cohort. Relative risks of prostate cancer were estimated by conditional logistic regression. RESULTS: Positive associations with prostate cancer risk were observed for choline and vitamin B2, and an inverse association was observed for MMA. The relative risks for a doubling in concentrations were 1.46 [95% confidence interval (95% CI), 1.04-2.05; P(trend) = 0.03] for choline, 1.11 (95% CI, 1.00-1.23; P(trend) = 0.04) for vitamin B2, and 0.78 (95% CI, 0.63-0.97; P(trend) = 0.03) for MMA. Concentrations of betaine, cysteine, methionine, and vitamin B6 were not associated with prostate cancer risk. CONCLUSION: The results of this large prospective study suggest that elevated plasma concentrations of choline and vitamin B2 may be associated with an increased risk of prostate cancer. These novel findings support a role of one-carbon metabolism in prostate cancer etiology and warrant further investigation.